Name: MUFE 22.08.2020
Sign: MATH2059 Midterm Exam Duration: 6±0.1 hr

1. (15 p) The Maclaurin series expansion for the arctangent of x is defined for $|x| \leq 1$ as:

$$\tan^{-1} x = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1}$$

Compute the number of terms needed to approximate the arctangent of x for $x = \frac{\pi}{6}$ that gives an error of no more than $10^{-10}$

2. (10 p) Convert the decimal number 15.796875 to the Hexadecimal, Octal and Binary numbers exactly.
3. (15 p) In 1225, Leonardo of Pisa (Fibonacci) found one of the roots of the following cubic equation $x^3 + 2x^2 + 10x = 20$ to be 10-digit accuracy. Nobody knows by what method he achieved that. Find this root to within ***16*** significant digits. How can you find the other roots easily? Briefly explain *and* find them. Write your own M-file based on any root finding method and explain the reason for your choice.
4. (15 p) Determine the roots of the simultaneous nonlinear equations $\begin{Bmatrix} y = -x^2 + x + 0.75 \\ y + 5xy = x^2 \end{Bmatrix}$. Use a graphical approach to obtain your initial guesses. Determine refined estimates with solver tool from Excel or a software package of your choice.
5. (15 p) Solve the following system of equations using Doolittle's LU-decomposition method with **partial pivoting** for solution (*Clearly express **lower** and **upper** triangular matrices*).

$$\begin{aligned} 2c_1 - 6c_2 - c_3 &= -38 \\ -3c_1 - c_2 + 7c_3 &= -34 \\ -8c_1 + c_2 - 2c_3 &= -40 \end{aligned}$$

6. (15 p) Solve previous problem with the Gauss-Seidel method to $\varepsilon_s = 10^{-4}\%$.
7. (15 p) The data below represents the bacterial growth in a liquid culture over a number of days.

| Day | 0 | 4 | 8 | 12 | 16 | 20 |
|---|---|---|---|---|---|---|
| Amount × $10^6$ | 67 | 84 | 98 | 125 | 149 | 185 |

Find a best-fit equation to the data trend. Try several possibilities-linear, parabolic, and exponential. Use the software package of your choice to find the best equation to predict the amount of bacteria after 40 days. Display the results graphically.

**Note**: **your answer sheet should involve script, result and your comments.** Good Luck